



BobSim

SteadyStateEval Vehicle Characterization

OpenModelica Executable with Curvature Feedforward and PI Trim

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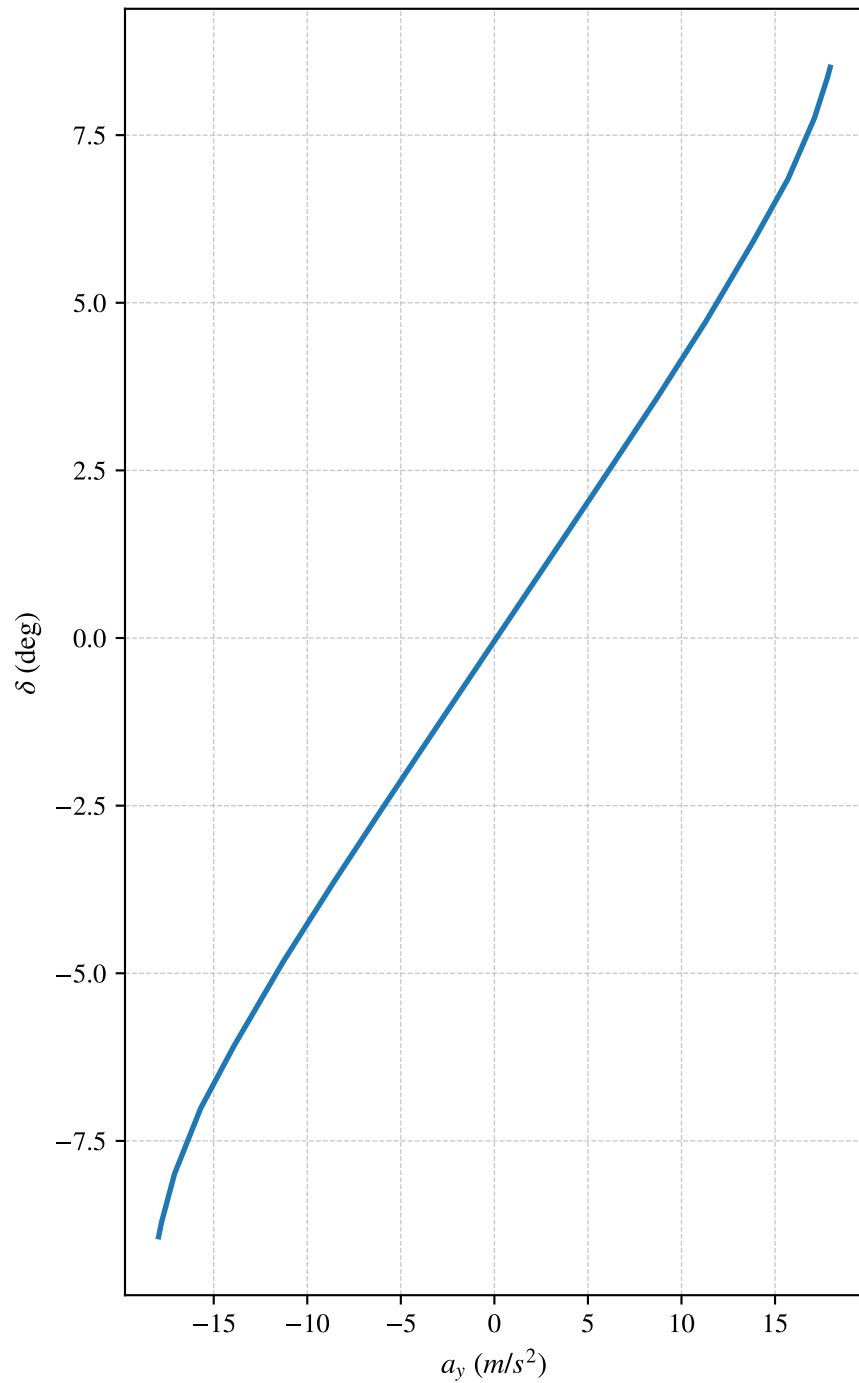
BobDyn

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- SteadyStateEval steady-state circular test performed
 - Constant velocity maintained at 15 m/s
 - Radius sweep from 12.5 m to 100 m
 - Tire model: MF52

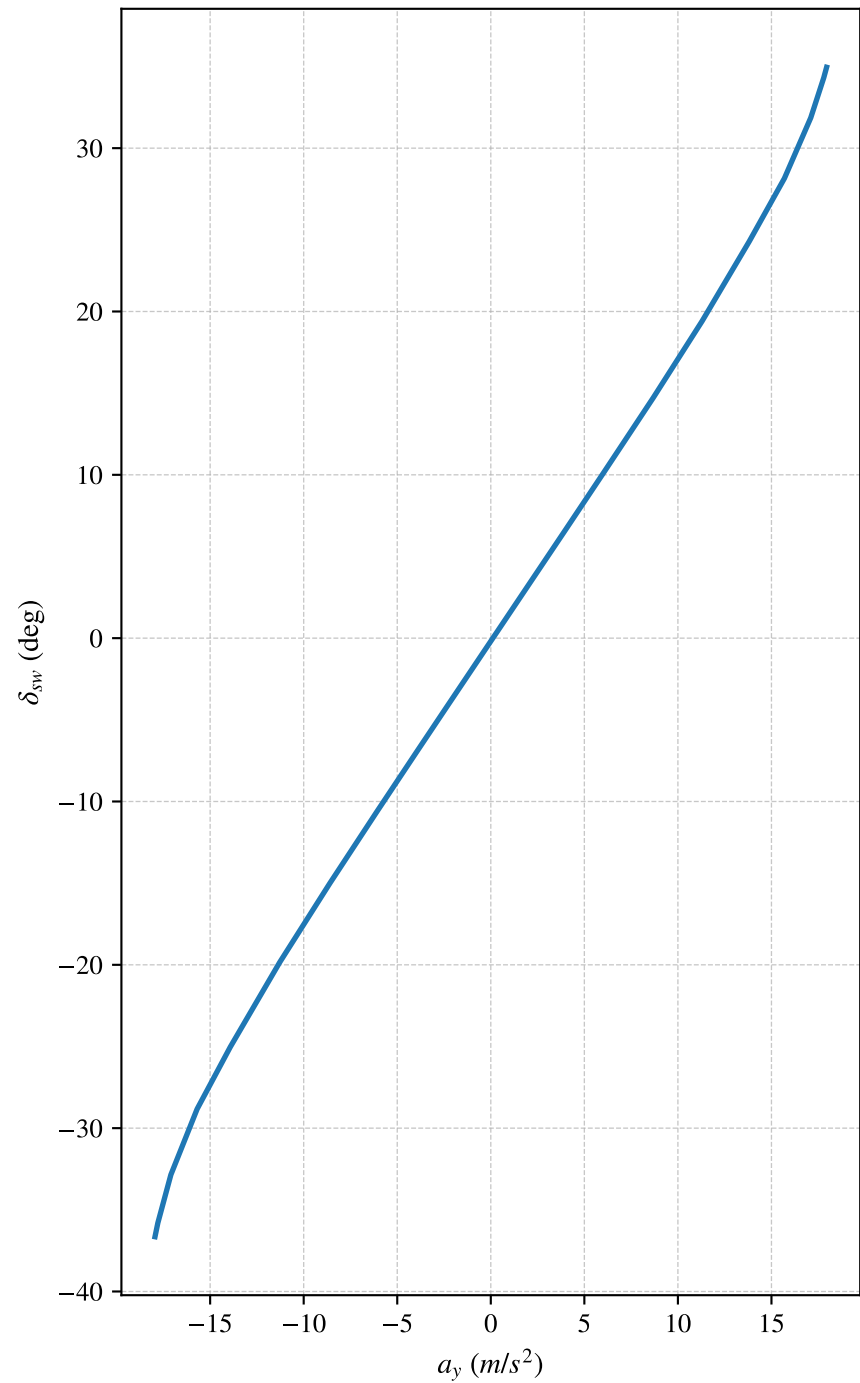
Metric	Value	Units
a_y Range	$-17.95 \rightarrow 17.95$	m/s^2
Roadwheel Understeer Gradient	0.00725	$\frac{\text{rad}}{\text{m/s}^2}$
	4.07	$\frac{\text{deg}}{g}$
Handwheel Understeer Gradient	0.02982	$\frac{\text{rad}}{\text{m/s}^2}$
	16.76	$\frac{\text{deg}}{g}$
Roll Gradient	1.257	$\frac{\text{deg}}{g}$
Max Radius Error	4.65	%

Steering Angle vs a_y

Roadwheel Angle δ

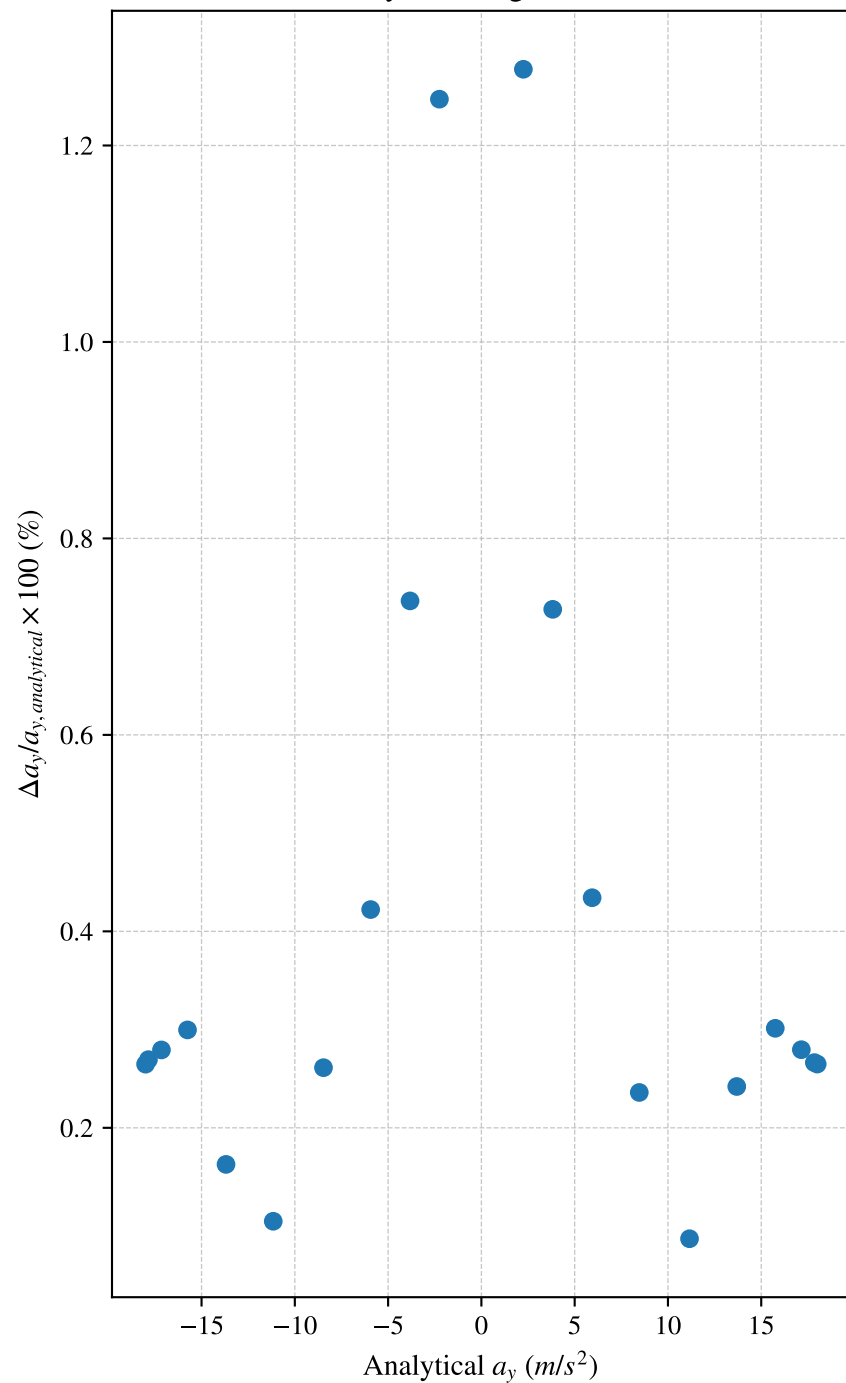


Handwheel Angle δ_{sw}

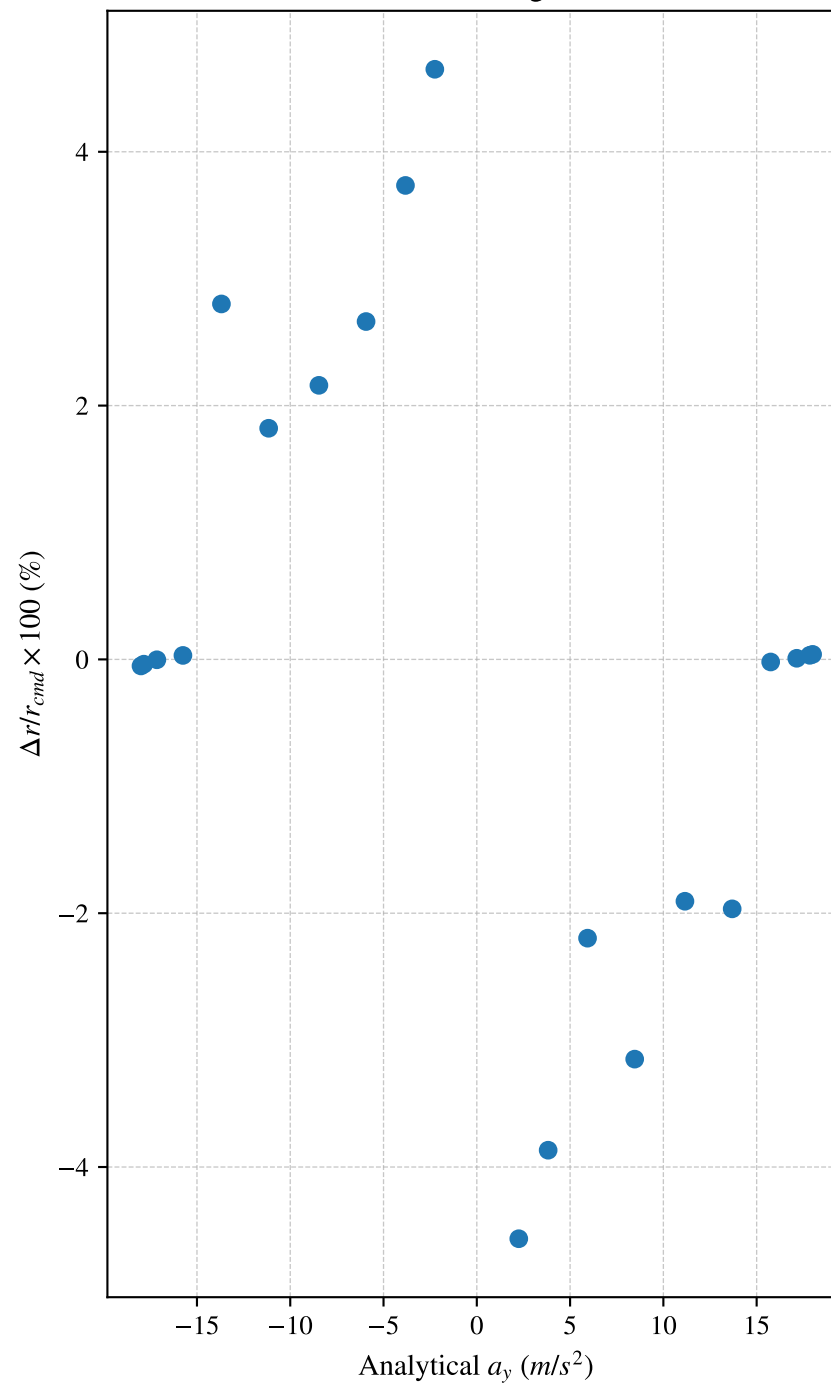


Tracking Error vs Analytical a_y

Ay Tracking Error

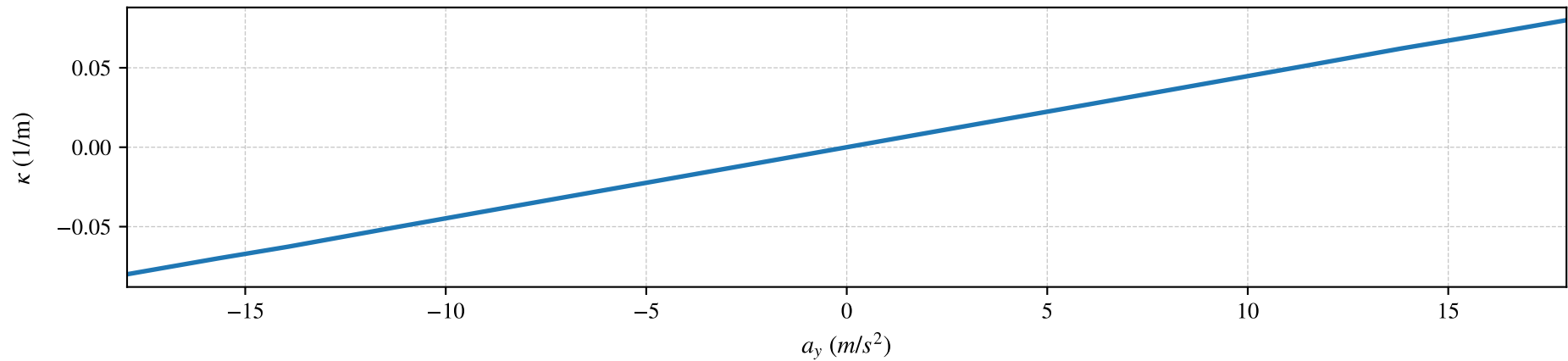


Radius Tracking Error

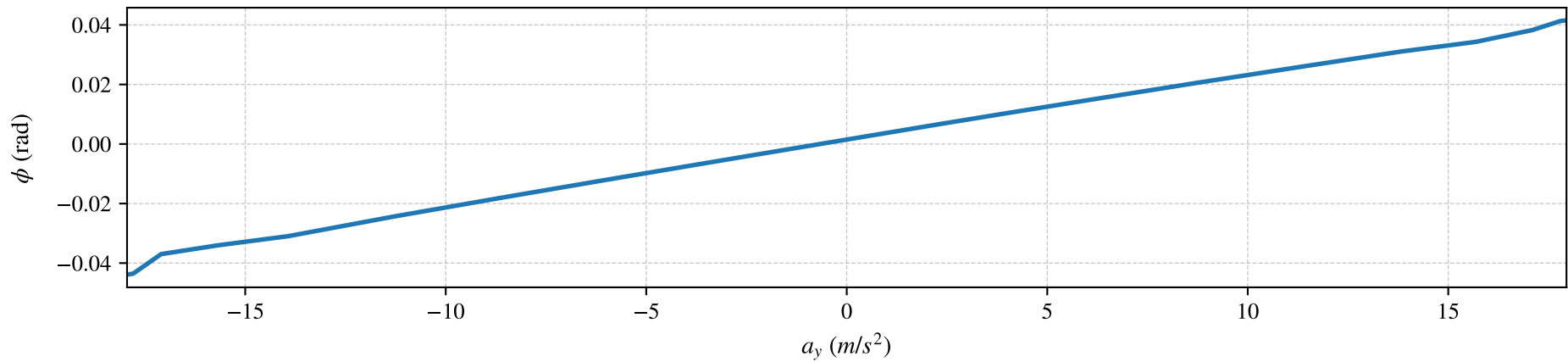


Vehicle States vs a_y

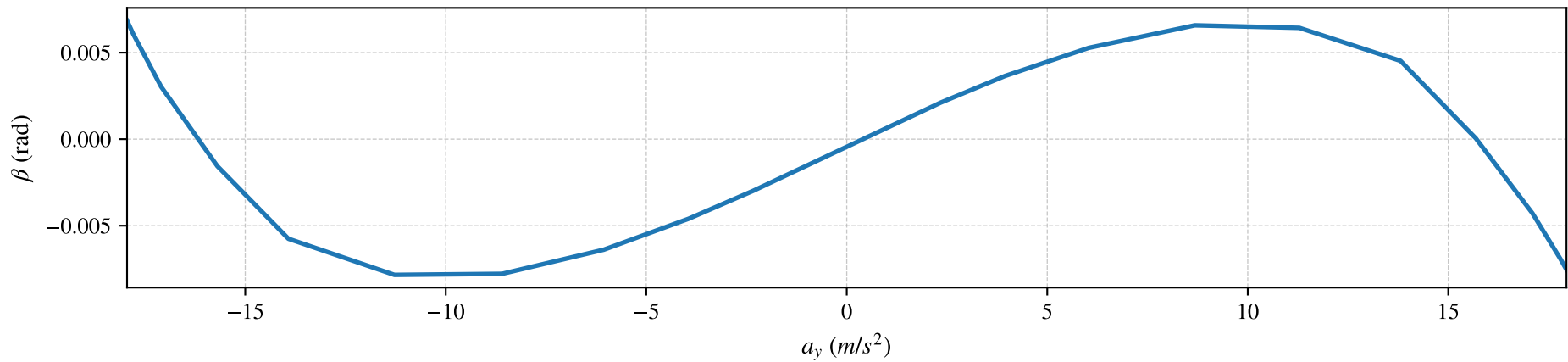
Curvature κ



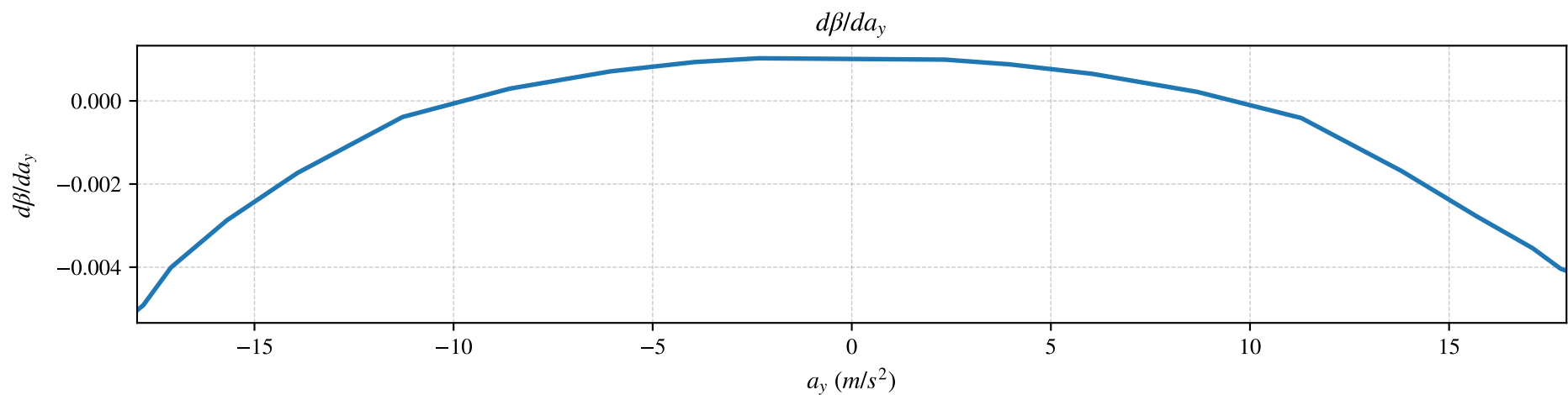
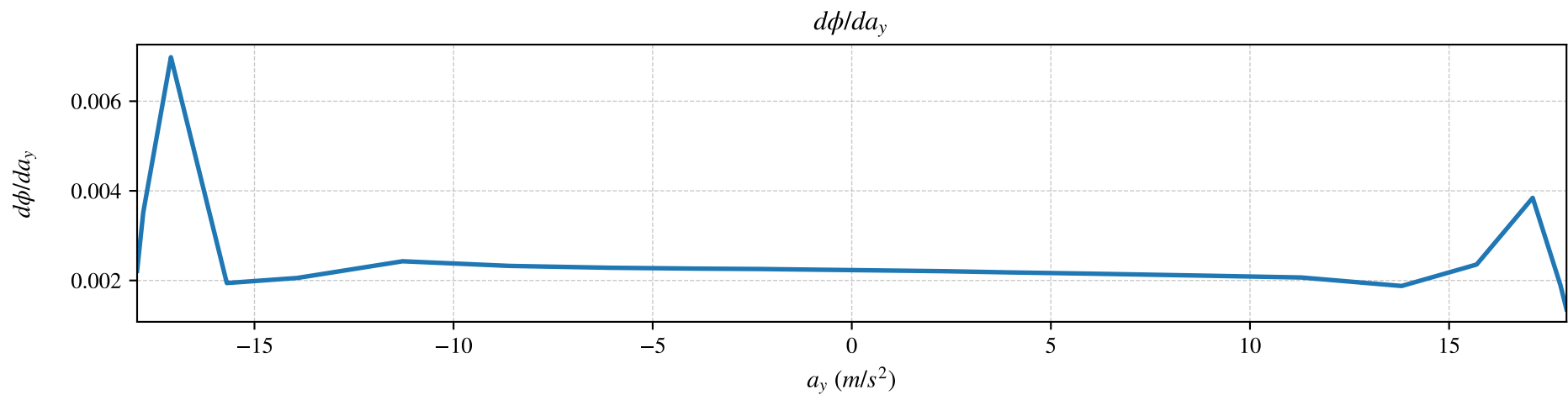
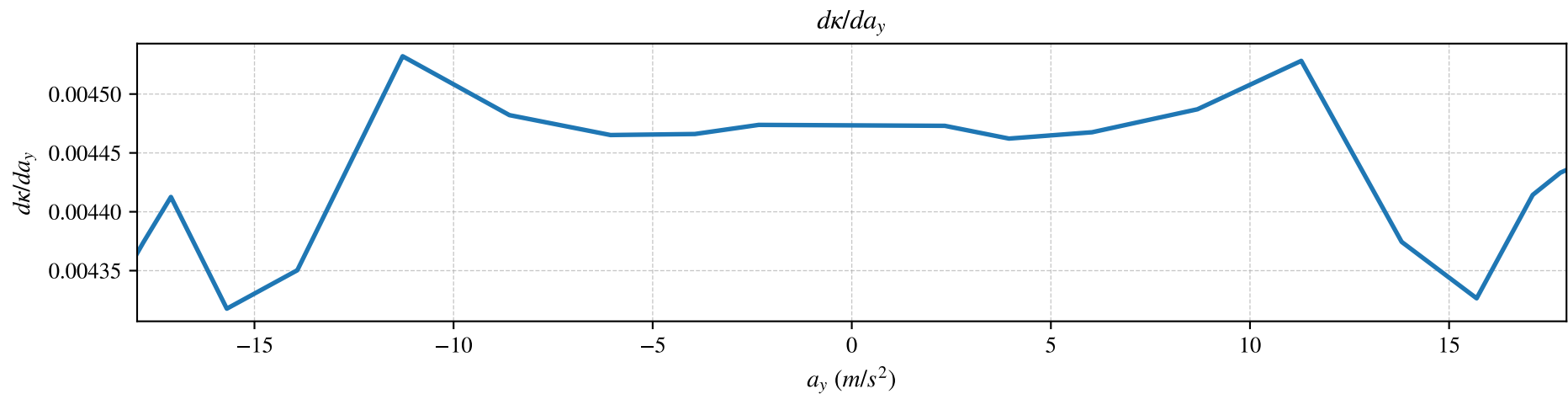
Roll Angle ϕ



Sideslip β

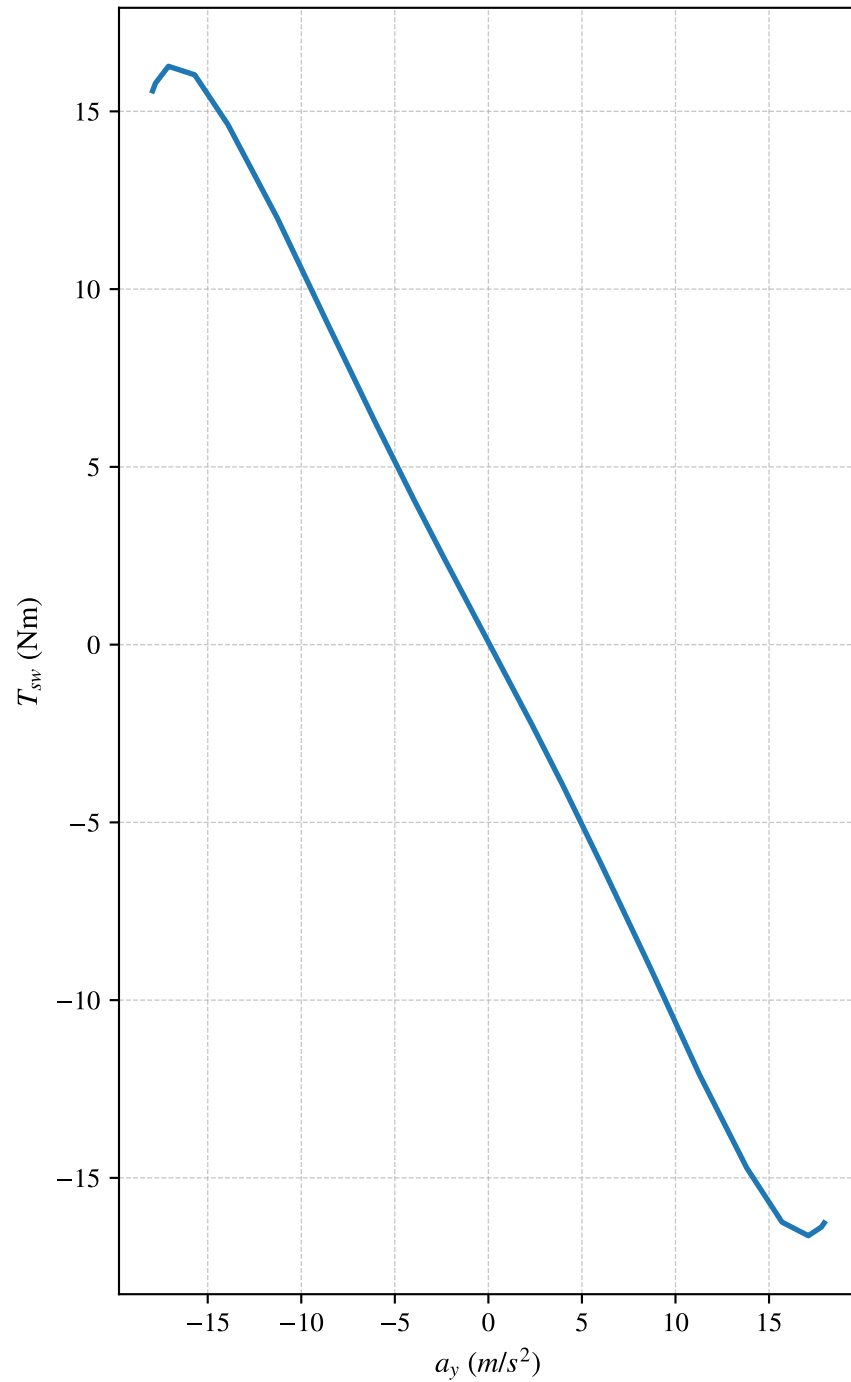


Response Sensitivities vs a_y



Steering Effort and Sensitivity vs a_y

Steering Torque T_{sw}



Steering Sensitivity $d\delta/da_y$

